Dual AUDIO LEVEL SHIFTER And BUFFER For ADCs, MCU Kits And Arduino

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Many single-ended analogue signals do not have DC components or appropriate DC offsets. Due to that, these require very high input resistance for analogue-to-digital converter (ADC) inputs. This simple universal dual level shifter and buffer for AC and audio signals can be used for ADCs, MCU kits and Arduino. It does not need any special adjustment to work properly. Before using the circuit, select and adjust the appropriate DC offset for each operational amplifier, every time.

SELECTION OF DC OFFSETS

Closed jumper	Function
SJ1	Offset Vcc/2 for A1 of IC1, here Vcc=5V
SJ2	Offset Vcc/2 for A2 of IC1, here Vcc=5V
SJ3	Adjustable DC offset with VR1 for A1 of IC1
SJ4	Adjustable DC offset with VR2 for A2 of IC1

We need analogue level shifters and buffers between the source of analogue signals (AC or audio) and inputs of the ADC. Here, we see a simple solution for two channels of the ADC. The circuit can be multiplied for any number of ADC inputs.

Circuit and working

Circuit diagram of the dual and audio level shifter and buffer for ADCs, MCU kits and Arduino is shown in Fig. 1.

The circuit has two analogue channels. In each channel, there is a single operational amplifier working as a level shifter and buffer follower. Input AC or audio signal is applied to input1 and input2. Input resistance at low frequency is around 1-mega-ohm. Bandwidth of each channel is 2Hz to more than 100kHz.

PARTS LIST

Semiconductors:	
IC1	- MCP602 op-amp
D1	- 1N4007 rectifier diode
D2-D8	- 1N4148 signal diode
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R4	- 300-ohm
R2, R3, R8, R9, R11	- 1-kilo-ohm
R5, R12	- 1-mega-ohm
R6, R10	- 100-ohm
R7, R13	- 100-kilo-ohm
VR1, VR2	- 10-kilo-ohm potmeter
Capacitors:	
C1	- 220µF, 16V electrolytic
C2-C4	- 100nF ceramic disk
C5-C7	- 100µF, 16V electrolytic
Miscellaneous:	
CON1	- USB type A connector
CON2, CON3	- 2-pin connector
CON4-CON7	- 2-pin berg strip
CON8, CON9	- 2-pin terminal connector
F1	- 100mA fuse with holder
SJ1-SJ4	- Shorting jumper

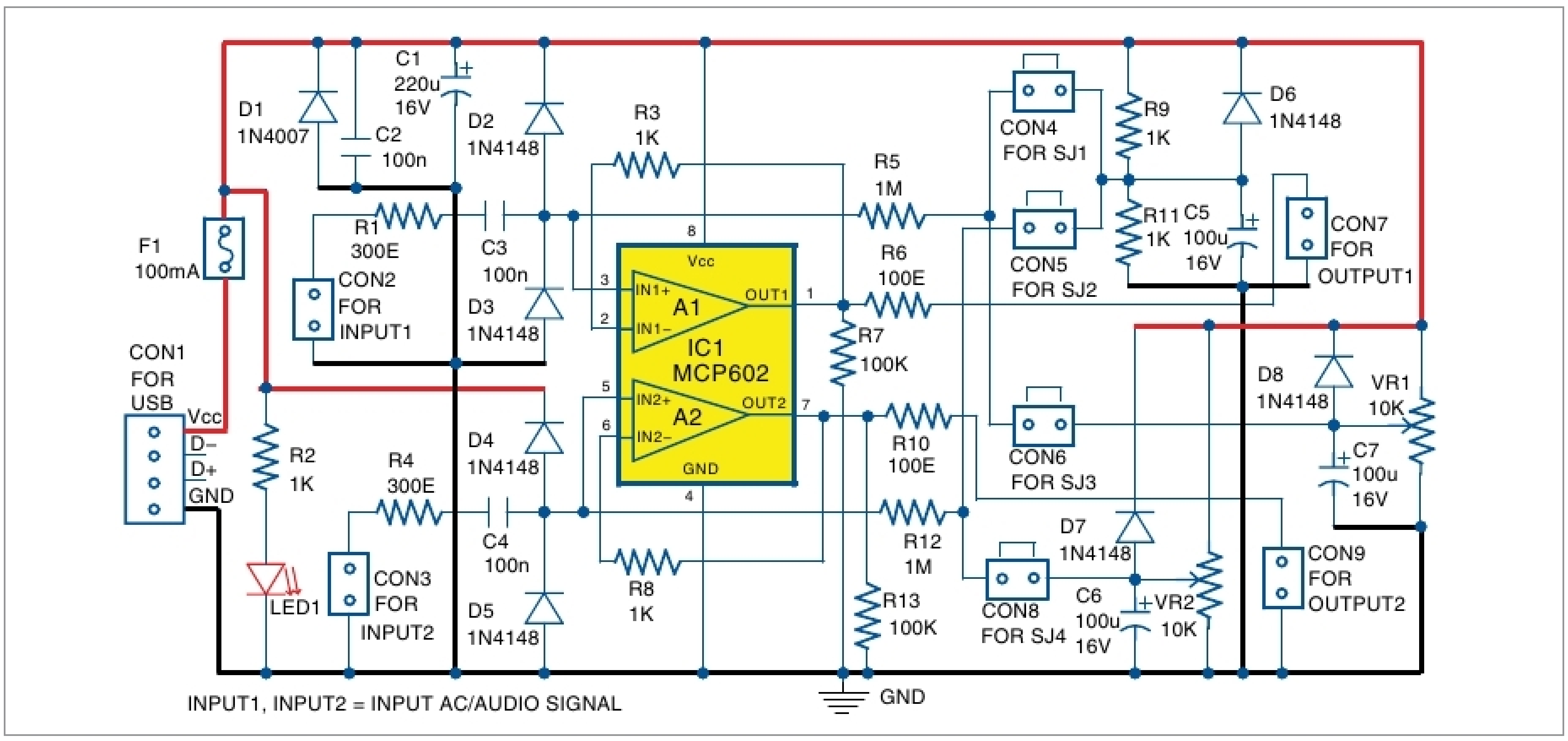

Fig. 1: Circuit diagram of dual level shifter and buffer